

NEWBERRY, SC, USA

WMO: 720596

Lat: 34.300N

Lon: 81.633W

Elev: 173

StdP: 99.26

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00189

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-2.6	-14.9	1.1	-1.6	-12.2	1.3	0.5	10.0	11.5	8.9	9.2	1.8	280	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	35.2	23.5	34.1	23.2	32.7	23.2	25.7	30.1	25.3	30.1	24.9	29.7	3.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.5	26.7	24.0	19.3	26.4	23.0	18.1	25.9	80.2	30.2	78.1	29.4	76.3	31.3	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.3	(b) 7.1	(c) 5.7	(e) -7.9	(f) 37.4	(g) 2.0	(h) 2.2	(i) -9.3	(j) 39.0	(k) -10.4	(l) 40.2	(m) -11.5	(n) 41.5	(o) -13.0	(p) 43.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	6.3	8.6	12.1	17.3	21.7	25.9	27.1	26.4	23.5	17.1	11.2	8.6
	DBStd	8.34	5.30	5.16	5.02	4.10	3.48	2.31	1.93	2.04	3.01	3.87	4.17	4.76
	HDD10.0	385	142	82	36	2	0	0	0	0	0	2	34	88
	HDD18.3	1525	374	274	202	68	14	0	0	0	3	68	217	304
	CDD10.0	3004	26	43	101	220	363	476	531	508	404	222	69	43
	CDD18.3	1105	1	2	8	36	119	226	272	249	157	30	2	2
	CDH23.3	10234	2	11	78	362	1012	2213	2738	2295	1258	253	9	2
CDH26.7	4354	0	0	9	78	356	1022	1322	1027	499	41	0	0	
Wind	WSAvg	2.3	2.7	2.9	2.8	2.8	2.2	2.1	2.0	1.8	2.0	2.3	2.2	2.4
Precipitation	PrecAvg	1150	116	98	118	77	82	106	97	113	102	82	75	83
	PrecMax	1493	361	189	296	177	158	276	212	433	226	243	264	214
	PrecMin	866	14	22	19	8	28	27	14	10	6	0	14	10
	PrecStd	175	58	45	55	46	37	61	52	75	54	67	49	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	25.1	27.8	30.9	34.8	36.4	37.3	36.2	34.1	29.9	24.9	22.7
		MCWB	17.2	17.3	16.7	18.6	21.4	23.2	23.7	22.7	22.2	20.6	17.9	18.3
	2%	DB	19.8	22.2	26.0	28.7	32.2	35.0	35.9	34.8	32.6	27.7	22.3	21.1
		MCWB	15.5	15.6	16.4	18.5	20.8	23.5	23.5	23.4	22.0	19.6	16.2	17.8
	5%	DB	17.4	19.9	22.8	27.4	30.3	32.9	34.2	32.8	31.3	26.2	20.9	18.0
		MCWB	13.8	15.0	14.9	18.2	20.7	23.2	23.6	23.7	22.1	18.3	15.6	16.2
	10%	DB	15.1	17.3	21.1	25.9	28.0	32.2	32.7	32.1	30.0	24.8	18.6	17.1
		MCWB	11.9	13.5	14.5	17.4	20.6	23.1	23.8	23.6	21.9	18.4	14.6	14.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	19.3	18.9	22.2	23.7	25.7	26.2	26.7	25.3	24.0	20.1	20.4
		MCDB	21.3	23.0	23.3	25.7	28.3	31.5	30.3	29.7	29.1	26.9	21.6	21.6
	2%	WB	17.1	17.7	17.9	20.9	22.8	25.1	25.6	25.8	24.3	22.8	18.7	18.5
		MCDB	18.5	19.6	22.3	23.8	27.7	31.0	30.6	29.6	28.6	24.3	21.3	19.9
	5%	WB	15.0	16.3	17.0	19.9	22.2	24.6	25.2	25.4	23.6	21.3	17.3	17.1
		MCDB	16.7	18.9	20.9	24.3	27.0	30.4	30.7	29.4	27.7	23.6	18.6	17.9
	10%	WB	12.4	14.2	15.9	18.7	21.7	24.0	24.8	24.9	23.0	19.7	15.9	15.2
		MCDB	14.1	16.9	19.6	24.4	26.0	29.5	30.1	29.1	27.0	23.0	18.0	17.2
Mean Daily Temperature Range	5% DB	MDBR	10.7	10.9	12.2	13.1	11.5	11.3	11.2	10.6	10.8	12.4	12.2	10.1
		MCDBR	13.3	12.6	14.6	14.4	13.6	12.9	13.1	12.2	12.9	14.2	13.9	11.1
		MCWBR	9.0	7.4	6.9	5.8	4.6	3.7	3.4	3.5	4.0	6.3	8.4	7.7
	5% WB	MCDBR	11.1	10.2	11.8	11.4	10.2	11.1	10.8	9.7	10.3	9.4	9.6	9.1
		MCWBR	9.1	7.4	6.9	5.6	4.0	3.8	3.8	3.6	3.8	5.3	7.3	7.3
Clear-Sky Solar Irradiance	taub	0.318	0.327	0.366	0.406	0.466	0.495	0.543	0.525	0.438	0.396	0.345	0.320	
	taud	2.497	2.470	2.391	2.267	2.145	2.107	1.994	2.034	2.258	2.347	2.450	2.525	
	Ebn at Noon	892	922	907	881	829	802	760	767	823	831	850	869	
	Edh at Noon	85	96	113	134	154	159	178	168	128	108	88	78	
All-Sky Solar Radiation	RadAvg	2.73	3.35	4.43	5.56	6.12	6.43	6.05	5.54	4.76	3.96	3.01	2.35	
	RadStd	0.14	0.37	0.32	0.42	0.49	0.46	0.40	0.27	0.42	0.47	0.25	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (58 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	+120

Nomenclature: See separate page